A Stochastic Machine Learning Surrogate Model for Comprehensive Replacement of Spray Submodels

ILASS-Americas 36th Annual Conference

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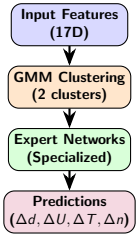

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Key Insight

Particle dynamics exhibits *learnable patterns* from state variables $\Rightarrow$ ML surrogates can replace expensive submodels

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- Single property cannot determine evolution $\Rightarrow$ **unsupervised multi-feature clustering**

Simulation Setup

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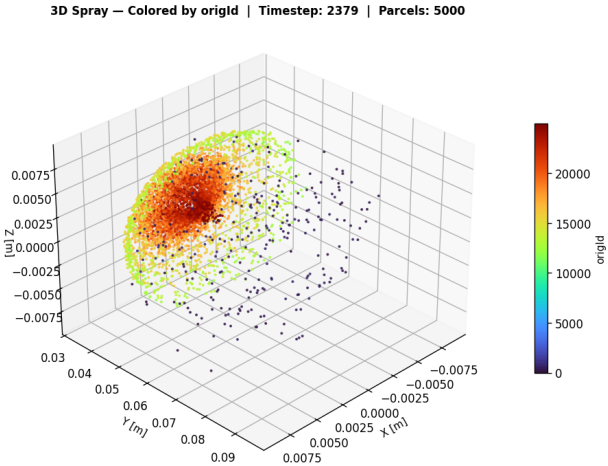
- Water spray: $\rho = 988$ kg/m^3, $\sigma = 0.0675$ N/m



3D spray colored by origId (25,000 parcels, mid-timestep)

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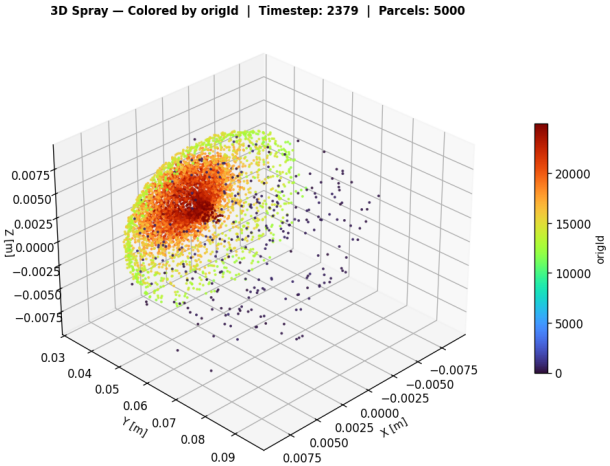

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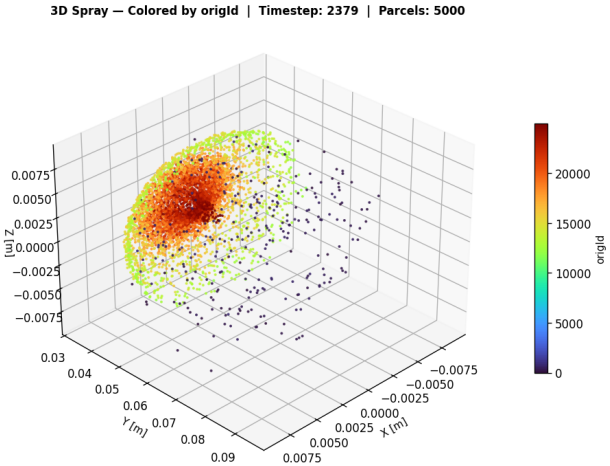

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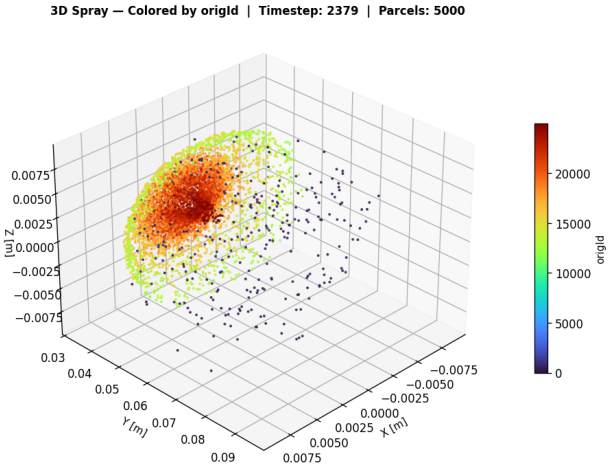

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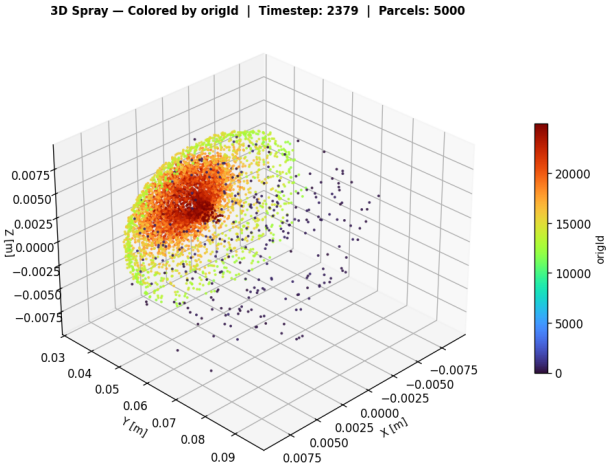
Dataset Statistics:

- Total pairs: 2,423,037
- Train/Val/Test: 80/10/10%
- Input: 17D (10 Lag. + 7 Eul.)
- Output: 6 targets



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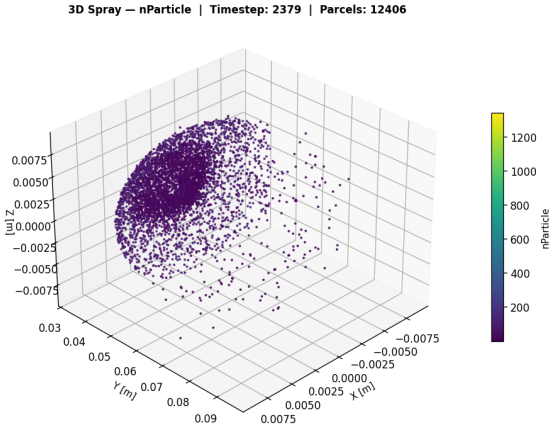
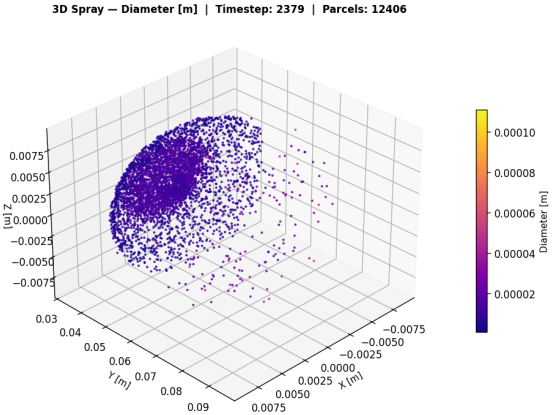
Spray Visualization: Particle Properties

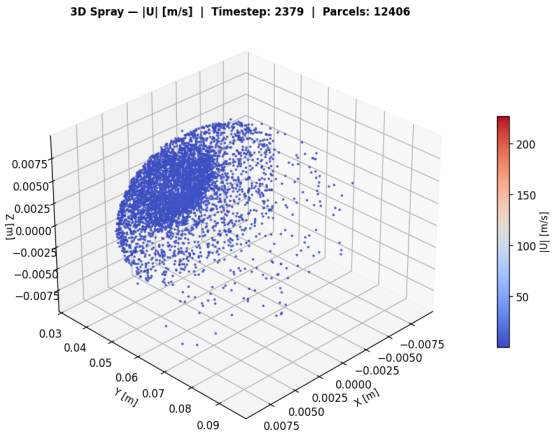


nParticle — Number of particles per parcel

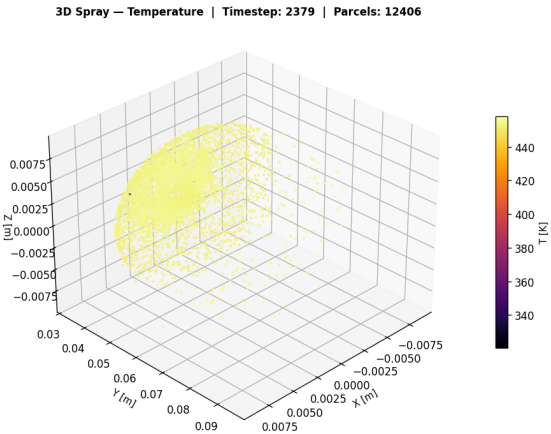


Diameter — Droplet size evolution

Spray Visualization: Flow Properties



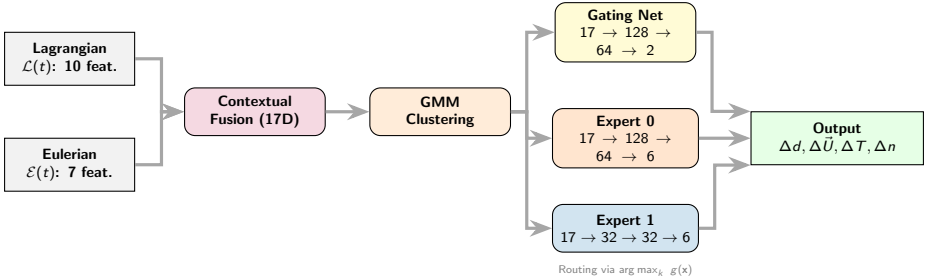
Velocity Magnitude — Momentum exchange with gas



Temperature — Heating from 323K ambient (800K)

Neural Network Architecture

SMLS Architecture: Standard Unsupervised Clustering (SUC)



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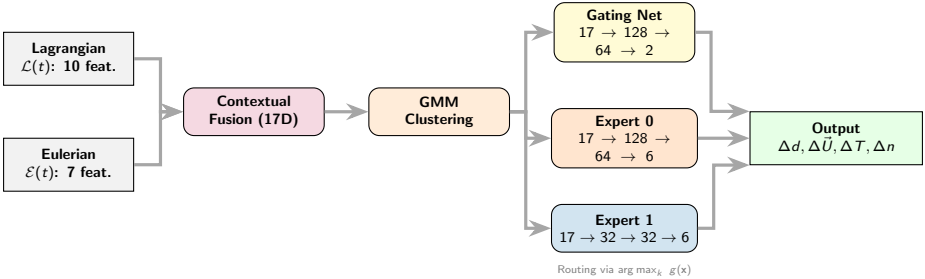

$$\Delta\hat{Y}_{\text{ensemble}} = \sum_{k=0}^{1} \mathbb{1}\left[k = \arg\max_{k} \; g(\mathbf{x}; \boldsymbol{\theta}_{gate})\right] \cdot f_k(\mathbf{x}; \boldsymbol{\theta}_k)$$

GMM Clustering

GMM Clustering: Physics-Informed Regime Discovery

Feature space for GMM:

$$\mathcal{G} = \{We,\ Re,\ \Delta d,\ \Delta T,\ \Delta n_{Particle},\ \Delta U_{rel,mag}\}$$

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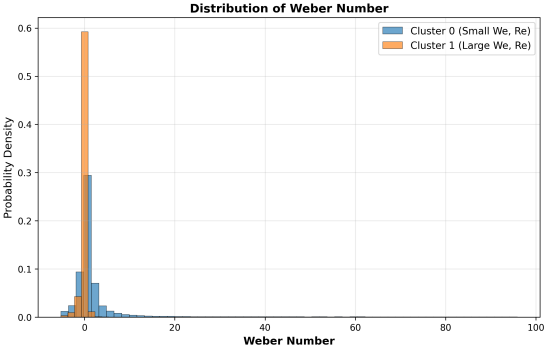
Key Non-Dimensional Numbers

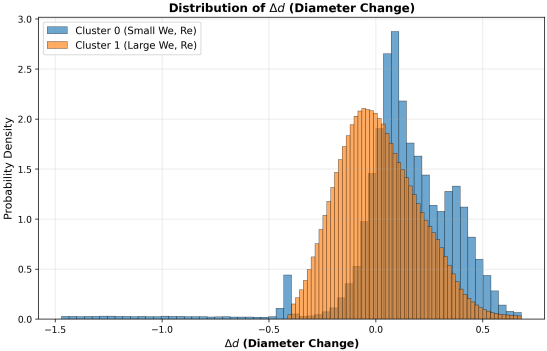
$$Re_p = \frac{\rho_g |\vec{U}_{rel}| d}{\mu_g} \qquad We = \frac{\rho_g |\vec{U}_{rel}|^2 d}{\sigma} \qquad U_{rel} = |\vec{U}_p - \vec{U}_g|$$

GMM Results: Non-Dimensional & Size Distributions

Weber & Reynolds Number

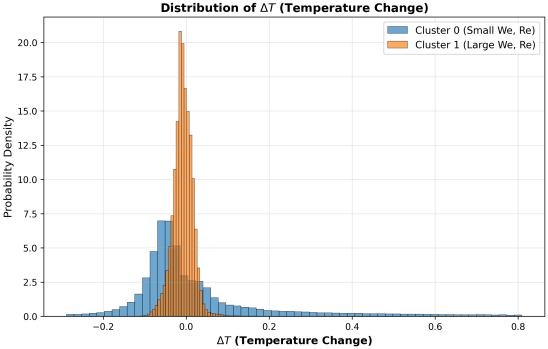


Diameter Change (Δd)

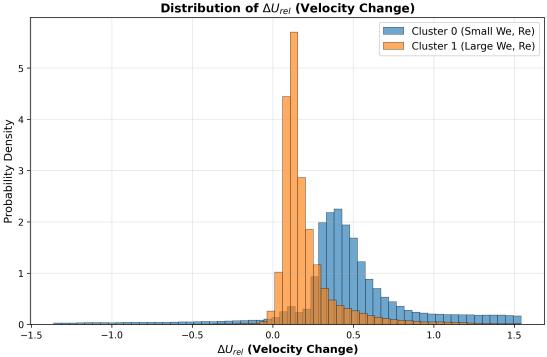


GMM Results: Thermal & Velocity Distributions

Temperature & Particle Change



Relative Velocity (U_{rel})



Supervised Learning

Supervised Learning: Gating + Expert Networks

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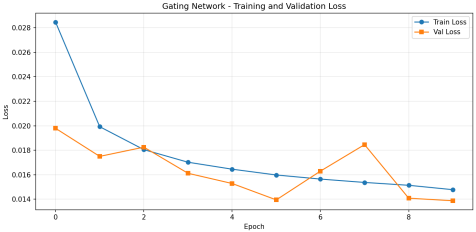
1. **Phase 1: Gating Network**
   - $17 \rightarrow 128 \rightarrow 64 \rightarrow 2$ (softmax)
   - Cross-entropy loss on GMM labels
   - **Accuracy: 99.42%** (10 epochs)



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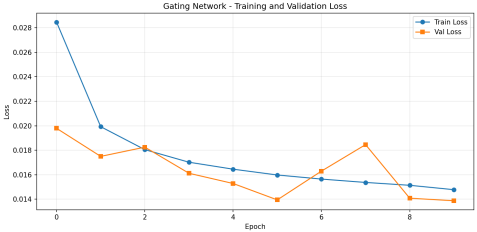
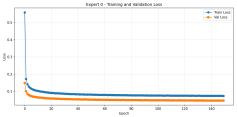
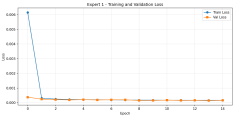
2. **Phase 2: Expert 0** (Small drops)
   - 17 → 128 → 64 → 6
   - 150 epochs, LR: 5e-5, batch: 64
   - **Test loss: 0.0447, R^2 = 0.9875**







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2. **Phase 2: Expert 0** (Small drops)
   - $17 \rightarrow 128 \rightarrow 64 \rightarrow 6$
   - 150 epochs, LR: 5e-5, batch: 64
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3. **Phase 3: Expert 1** (Large drops)
   - $17 \rightarrow 32 \rightarrow 32 \rightarrow 6$
   - 15 epochs, LR: 1e-3, batch: 256
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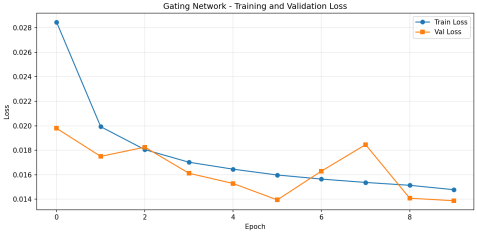

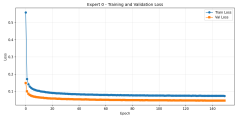

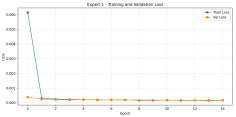

Supervised Learning: Gating + Expert Networks

Three-Phase Training Pipeline:

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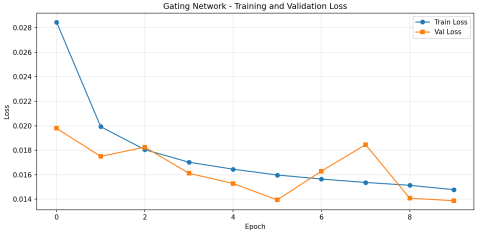
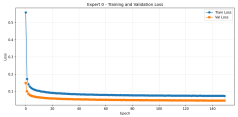
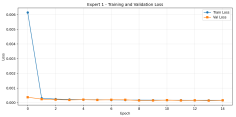




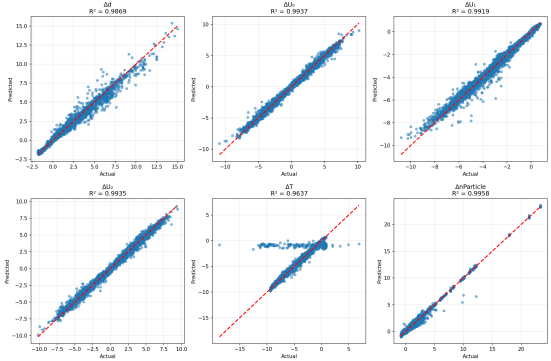


Loss Functions

Results

Results: Prediction Performance

Expert 0: Small/Volatile Drops (R^2=0.9875)



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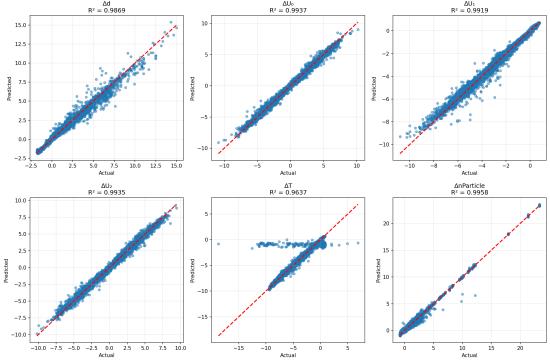

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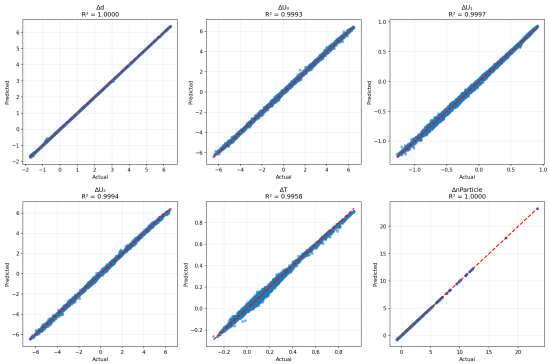

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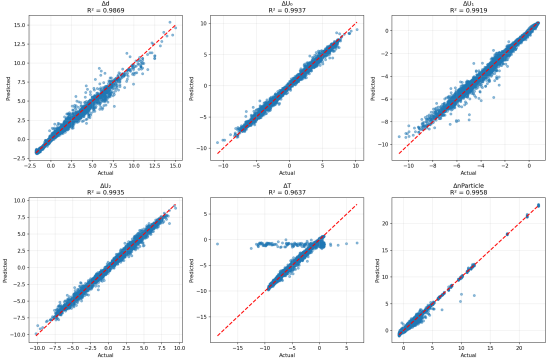

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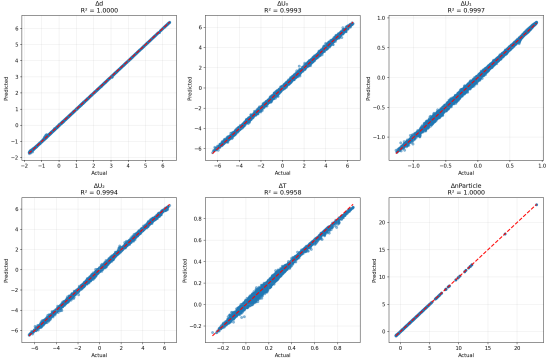

Ensemble R^2

$$R^2_{\text{ens}} = 0.137 \times 0.9875 + 0.863 \times 0.9990 = \mathbf{0.9974}$$

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  Hierarchical regime discovery (primary vs. secondary breakup, evaporation vs. drag dominated)

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- **A-posteriori Validation**
  Full coupled simulation with ML surrogate replacing submodels

Questions & Discussion

Questions?

Rohit Mishra
rmishra@tamu.edu

Code: `github.com/ctftamu/hybridUnsupervisedClustering`